

AssistEditor: Multi-Agent Collaboration for GUI Workflow Automation in Video Creation

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Abstract

Graphical User Interface (GUI) Automation has shown significant potential recently. Previous works built GUI Agent systems to handle short-procedure tasks such as element grounding or functional assistance. In this paper, we propose a novel PC-Copilot, AssistEditor, that focuses on automating the video editing workflow. Unlike previous approaches, our system does not require users to input specific commands to control the computer. Instead, users simply describe their requirements, such as the content and style of the video, and upload the necessary materials. The system then autonomously translates these requirements into detailed actions for controlling video understanding models and professional video editing software, e.g., Premiere Pro to produce the final video. This functionality is enabled by a collaborative AI agent framework of multiple GUI agents, each capable of dialogue, knowledge retrieval, and software usage. These agents have distinct roles, including interacting with users to gather requirements, generating storyboards, and performing editing tasks. This approach significantly streamlines the video editing process, making advanced editing accessible to users with varying levels of expertise.

CCS Concepts

• Computing methodologies → Artificial intelligence.

Keywords

Multi-agent Collaboration, GUI Automation, Video Editing

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1 Introduction

The potential for Graphical User Interface (GUI) Automation has been widely recognized in various fields, primarily for its ability to handle specific tasks such as element grounding and functional assistance. With the advent of Large Language Models (LLMs) [9, 11] and Multimodal Language Models (MLLMs) [1, 8], understanding and automating GUIs have become feasible and have recently garnered significant attention. Previous work [2, 12, 16, 18] mainly focused on understanding GUI elements, such as "Where should I click if I want to open a new file in a PC application?" or implementing specific short-procedure functions, like "Adding a blur effect in Adobe Premiere Pro." These approaches are suitable for isolated tasks but fall short when addressing more integrated workflows.

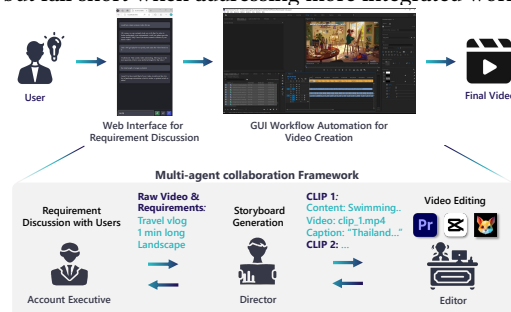


Figure 1: Diagram Illustration of AssistEditor for Workflow Automation.

We introduce, **AssistEditor**, a Multi-Agent Collaboration framework that leverages GUI Agents to achieve comprehensive workflow-level automation, as shown in Figure 1. Our primary demonstration focuses on the video editing workflow, a complex process that typically requires substantial manual effort and expertise. Our system significantly differs from existing solutions [3, 14, 15, 17] by eliminating the need for users to input detailed commands to control the computer. Instead, users are only required to describe their high-level requirements, such as the desired content and style of the video, and provide the necessary raw materials. The proposed model then autonomously converts these high-level requirements into specific actions within professional video editing software, invokes video understanding models, performs the necessary editing tasks, and ultimately generates the final video.

The core of our system is a multi-agent collaborative framework [6]. Each AI agent within this framework is capable of dialogue, knowledge retrieval, and software manipulation. These

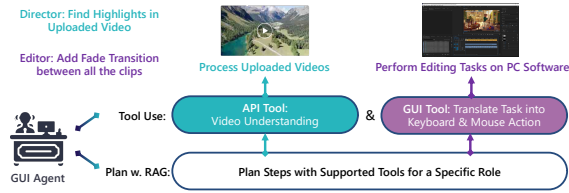


Figure 2: Diagram Illustration of GUI Agent.

agents are assigned distinct roles, including interacting with users to gather and refine their requirements, generating storyboards, and executing video editing tasks. This collaborative approach ensures that complex workflows can be managed efficiently and effectively, making sophisticated video editing accessible to users regardless of their technical skill level.

2 Technology

2.1 Overview of AssistEditor

Our system is designed for dialogue-based workflow automation in video editing. The functionality of the system is as follows:

Input: The user communicates their high-level requirements through a dialogue interface. This includes details about the desired content, style, and other specific elements they want in the video. Additionally, the user may optionally upload the necessary video materials, such as video clips, images, and audio files, that will be used in the editing process.

Output: The system automatically processes the user’s input and performs the video editing tasks according to the specified requirements. The final output is the edited video that aligns with the user’s instructions, which is then returned to the user.

2.2 Multi-Agent Collaboration Framework

The system is driven by a multi-agent collaboration framework based on GUI Agents, designed to mimic the roles and responsibilities commonly found in video production. This division of labor ensures that complex video editing tasks are handled efficiently and effectively. As illustrated in Figure 1, the framework consists of three primary roles: *Account Executive*, *Director*, and *Editor*.

Account Executive (AE). The AE is responsible for interacting with the user to understand their requirements. Operating in an active information-seeking mode, the AE is prompted to ask the user targeted questions, clarifying key details necessary for the editing process. Once the interaction with the user is complete, the AE automatically summarizes the conversation, providing a comprehensive brief for the Director to proceed with the next steps.

Director. The Director’s role is to draft a creative plan or storyboard based on the user’s requirements and the uploaded video materials. By utilizing the video temporal grounding pretrained model – UniVTG [7], the Director analyzes the video content to identify highlights and determine the optimal sequence for presenting various clips. The Director organizes this information into a storyboard format, ensuring a coherent narrative flow.

Editor. The Editor’s task is to translate the storyboard into a concrete editing plan and execute the necessary operations within the video editing software. This involves performing precise editing actions, such as trimming clips, applying effects, and adjusting audio, to produce the final video.

This multi-agent collaboration framework streamlines video editing, leveraging specialized roles to deliver high-quality outputs with minimal user intervention.

2.3 GUI Agent

Unlike previous work, such as AutoGPT [13], Devin [4], where each agent in a multi-agent framework is typically a language-based agent, AssistEditor is unique in that each agent is multimodal and additionally equipped with the ability to use software in a human-like manner, capable of performing GUI tasks, as shown in Figure 2. Thus, the architecture of each agent includes a knowledge base (KB) [10] for planning with retrieval-augmented generation (RAG) [5], API call capabilities, and GUI execution abilities.

Each role within the system is associated with different elements. For example, the **Account Executive (AE)**’s goal is to understand the user’s requirements. The AE’s knowledge base is centered around basic requirements for video creation and does not require API call or GUI execution capabilities.

For the **Director**, the goal is to generate a storyboard. The Director’s knowledge base focuses on the rules and specific functions for generating storyboards, video understanding APIs, and the GUI capabilities necessary for sourcing proper media.

The **Editor**’s goal is to translate the storyboard into an actionable editing plan and execute it within the video editing software. The Editor is equipped with a knowledge base about editing techniques, relevant API calls for video processing, and GUI capabilities to perform detailed editing tasks, e.g., create a project, import media files, add captions, etc.

This integration of GUI capabilities within each agent allows the system to perform complex video editing tasks more effectively, leveraging direct software manipulation to achieve high-quality outcomes with minimal user intervention.

2.4 Online Demo

The demo system is divided into three main components: the web interface, the PC with Windows environment, and the server.

The **web interface** facilitates the interaction between the user and the AI agents. It includes a chatbox for exchanging requirements and uploading video materials.

The **PC** is for GUI agents to operate software and complete specific tasks. Users can observe the AI’s operations on the software, giving them insight into how the video is being edited in real-time.

The **server** is the core where the primary decision-making processes of the AI agents occur. It can remotely control the PC to perform specific editing tasks.

The Video Creation service can be accessed through the Web browser. The user interacts with the AI through a chatbox on the web interface. Then, users can observe the AI’s software operations on the PC, providing transparency into how the tasks are being performed. Once all editing tasks are completed, the user can download the final video through a specified URL.

3 Conclusion

We propose AssistEditor, a multi-agent collaboration framework that boosts GUI automation for video editing at the workflow level. Integrating intelligent agents for dialogue, knowledge retrieval, and software manipulation, it enhances efficiency and makes video editing accessible to novice users.

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